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Project proposal

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Which domain and the problem you will address?

There are four domains:

- The home domain with rooms and floors.
- The comfort system domain with things such as lamps, blinds and sensors.
- The energy system domain with solar panels and batteries.
- The Security system domain with security cameras, sensors and sprinklers.

The problem to be addressed is that these systems often are setup as three separate systems. The project will unify these three system within a smart home system to one model where they can communicate across the systems to improve the effectiveness of the system. Some examples that the unifying the systems will improve:

- The lamps dim could be changed depending on battery level and the HVAC system could go into eco mode.
- The systems could communicate with no need for several temperature sensors separately.

What models and metamodels are needed?

There will be 5 metamodels needed in this project this include:

- HomeMM that represents the house layout.
- ComfortSystemMM that represents the comfort system in the home including things like lamps, HVAC and more.
- EnergySystemMM that represents the energy system in the home including this like solar panel, batteries and more.
- SecuritySystemMM that represents the security system including things like fire alarms, security cameras and more.
- SmartHomeSystemMM that interacts with all the other metamodels that should include the full smart home system including the different independent systems and the home layout.

How many domains, MMs, and their relations.

Four domains:

- Home domain that provides the physical structure
- Comfort domain that controls the environment and comfort parts in the home
- Energy domain that manages energy creation and handling.
- Security domain monitors the home safety including burglar systems and fire alarms.

The HomeMM is in charge of the home domain, the ComfortSystemMM is in charge of the comfort domain, the EnergySystemMM is in charge of the energy domain and the SecuritySystemMM is in charge of the security domain.